

Development Dynamics in 6 African Nations

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Abstract—

This report explores the multifaceted dimensions of development across six African countries using World Bank indicators from 2013 to 2023. Through five thematic lenses—economic resilience, education, health and wealth, technology and trade, and governance quality—the analysis uncovers patterns and disparities shaping national progress. Data visualizations reveal that while increased investment in health and education often correlates with improved outcomes, institutional quality and policy implementation play a critical role in driving sustainable development. The findings emphasize the need for integrated, evidence-based strategies that align economic growth with social equity and strong governance.

1. Introduction

Understanding the drivers of development is critical for shaping effective policies in Africa. This report presents a data-driven examination of six African countries over the period 2013 to 2023, drawing on key indicators from the World Bank. The analysis is organized around five themes: Economic Resilience, Education, Health and Wealth, Technology and Trade, and Governance & Institutional Quality. Each theme is explored through interactive visualizations and targeted comparisons to highlight progress, disparities, and structural challenges. By combining empirical evidence with contextual insights, the report aims to provide a nuanced view of development dynamics and inform strategies for sustainable growth and inclusive prosperity across the continent.

2. THEME 1: Economic Resilience & Fragility

This theme explores how selected African economies have responded to systemic shocks between 2013 and 2023, using GDP per capita, unemployment, remittances, and inflation as key indicators.

Figure 1 illustrates the GDP per capita trends from 2013 to 2023 across selected African countries. Notably, Morocco, Ghana, and Rwanda exhibit a strong post-COVID recovery, marked by a consistent upward trajectory in GDP per capita since 2020. This suggests that despite the initial shock, these countries have demonstrated significant economic resilience—potentially driven by adaptive policy responses, increased domestic productivity, and diversified economies.

Figure 2 compares GDP per capita against remittances as a percentage of GDP. A key insight is the stabilizing role of remittances, particularly during downturns. Countries with higher remittance inflows appeared less volatile in GDP trends, reinforcing the idea that remittances serve as an informal financial safety net, cushioning households and economies when traditional systems are under stress.

Figure 3, a correlation heatmap, reveals an intriguing relationship: GDP and unemployment are positively correlated (0.8), an atypical pattern. This could reflect serious income inequality, post-crisis structural mismatches in labor markets, where growth may be driven by capital-intensive sectors or recovery has not yet translated into job creation. It underscores that GDP growth alone does not guarantee inclusive recovery.

3. Theme 2: Education & Human Capital

This theme explores the relationship between educational investment, literacy, school completion rates, and unemployment. Nigeria stands out as the only country whose education expenditure has consistently remained below 1% of GDP (**Figure 4**). Despite this, it surprisingly leads in secondary school completion rates, indicating that budget increases alone may not guarantee improved educational outcomes (**Figure 5**).

Figure 5 showing a comparison of school completion rates reveals that while most countries lag in secondary education, Nige-

GDP per Capita (2013–2023)

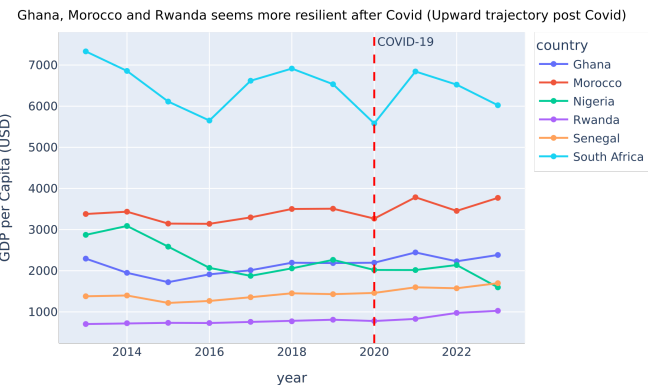


Figure 1. GDP per Capita for Selected Countries (2013–2023).

GDP per Capita vs Remittances (2013–2023)

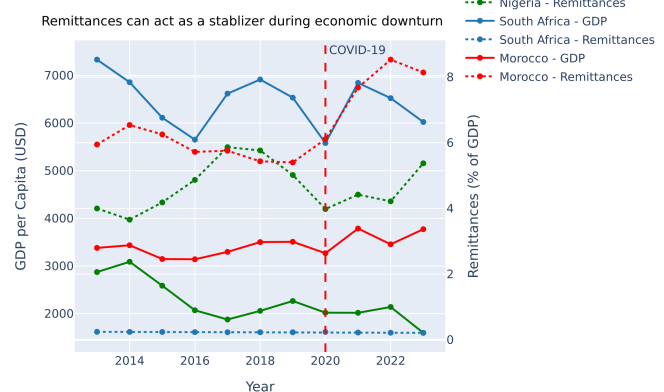


Figure 2. GDP per Capita vs. Remittances as % of GDP (2013–2023).

Correlation Heatmap – Economic Indicators

Positive correlation between GDP and Unemployment points to serious income inequality in these countries!!!!!!

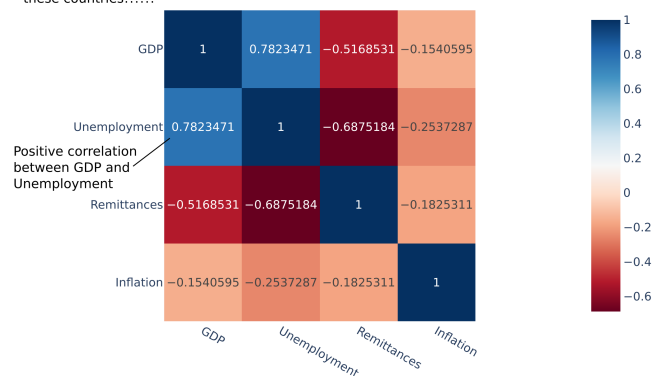


Figure 3. Correlation Heatmap of Development Indicators.

ria leads, followed by South Africa and Ghana. Interestingly, Nigeria and Ghana also report lower unemployment rates, suggesting a possible link between education outcomes and labor market absorption (Figure 6). However, South Africa presents an exception, with the highest unemployment rate despite high education completion—highlighting deeper structural challenges beyond education alone.

Together, these insights emphasize that while education funding is crucial, complementary policies and systemic reforms are equally essential to translating investment into human capital development.

Government Expenditure on Education (% of GDP)

Nigeria's Education Expenditure has consistently been below 1 percent of their GDP
How does this affect their human capital?

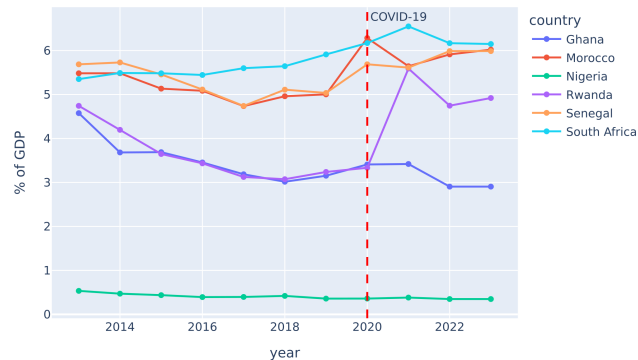


Figure 4. Government education expenditures for Selected Countries (2013–2023).

Education Completion Rate, population 25+ (cumulative) (2023)

Nigeria surprisingly has a very high primary and secondary school completion rate:
Maybe Governments has to do more than just increasing their education budget!

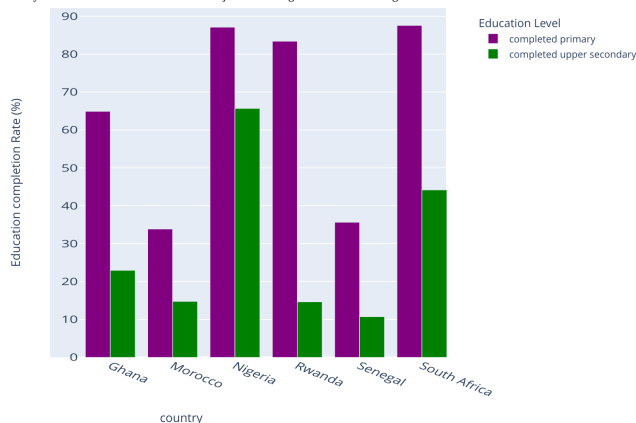


Figure 5. Education completion rate (2023).

4. Theme 3: Health vs. Wealth

This theme explores the relationship between a country's economic status and health outcomes. **Figure 7**, a bubble chart, plots GDP per Capita against Life Expectancy for six African countries in 2023, with bubble size representing current health expenditure (% of GDP). A clear trend emerges: countries with higher health spending generally exhibit longer life expectancy. However, economic wealth alone does not fully explain health outcomes—external factors such as political instability, violence, and access to quality care may also significantly impact life expectancy.

Figure 8, a multi-line chart for Morocco (2013–2023), compares trends in health expenditure and infant mortality. As health investment increased over the decade, infant mortality showed a consistent

Unemployment Rate and Government Education Expenditure (% of GDP) – 2023
Nigeria and Ghana are two countries with high secondary school completion rates and both have lower unemployment rate, South Africa appears to be an outlier.

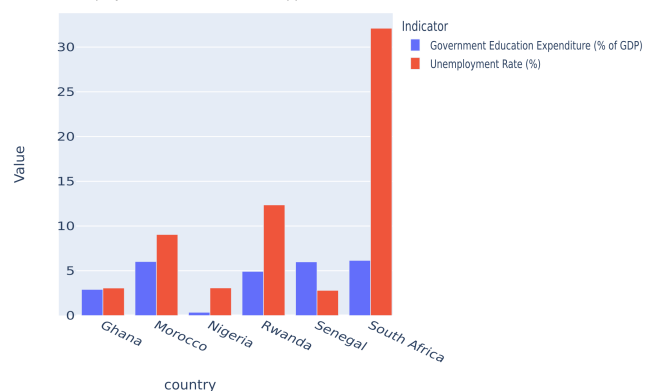


Figure 6. Unemployment and education expenditure comparison.

decline. This inverse relationship highlights the positive impact of sustained health funding on reducing preventable deaths, especially in early childhood.

Together, these charts underscore the importance of balanced investments in both economic development and healthcare infrastructure to improve public health outcomes across the continent.

Health vs. Wealth (2023)

The size of the bubbles denotes the government's health expenditure (% GDP):
Countries spending more on health care tend to have higher life expectancy.
Other factors such as crime and terrorism could also be playing a significant role in life expectancy

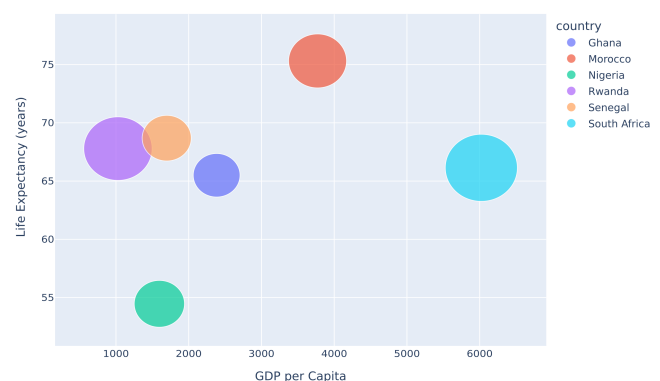


Figure 7. Health vs. Wealth (2023).

5. Theme 4: Technology, Trade & Education

This theme examines how education, digital connectivity, and trade relate to technological development across African countries.

In Nigeria, a multi-line chart comparing secondary school completion rates and internet access (2013–2023) reveals a strong parallel trend—as more individuals complete secondary education, internet usage rises. This suggests that educational attainment may play a key role in fostering digital inclusion **Figure 9**.

Figure 10, a bubble chart for 2023, plots Trade (% of GDP) against High-tech Exports (% of GDP), with bubble size representing Internet Usage. It highlights that countries with higher levels of high-tech exports tend to have greater internet penetration, while high trade volumes do not necessarily translate into high-tech export capacity. This underscores the distinction between trade openness and technological sophistication.

Finally, an area chart for Nigeria from 2013 to 2023 compares Agriculture, Industry, and High-tech Exports as contributions to GDP. The chart shows that tech-related exports remain minimal compared

Health Spending vs. Child Mortality – Morocco
Infant mortality rate declines as Health expenditure increases

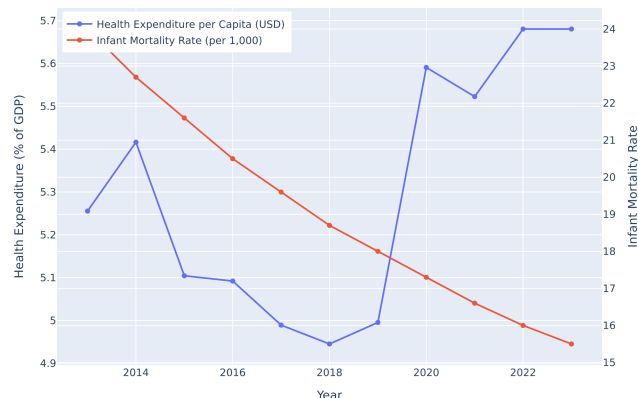


Figure 8. Health Spending vs. Child Mortality – Morocco.

to traditional sectors, reflecting Nigeria's continued reliance on agriculture and industry, and pointing to untapped potential in its digital economy **Figure 11**.

These insights emphasize the importance of linking education and infrastructure to technological advancement, while also recognizing that increased trade does not automatically drive tech growth without strategic investment.

Education vs Internet Usage in Nigeria

As the number of people completing secondary school increases, the percentage of the population with access to internet also increases.

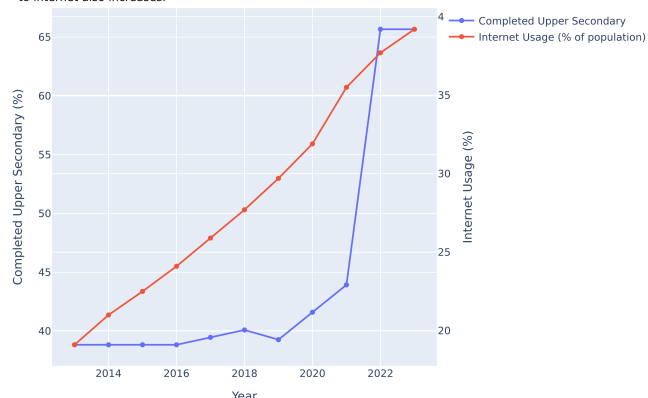


Figure 9. Education vs Internet Access in Nigeria.

6. THEME 5: Governance & Institutional Quality

This theme investigates how governance standards and institutional quality relate to a country's attractiveness for investment, especially foreign direct investment (FDI).

The radar chart (**Figure 12**) displays the performance of selected African countries across five governance indicators in 2023: Political Stability, Rule of Law, Voice and Accountability, Government Effectiveness, and Control of Corruption. Countries like Rwanda, Morocco and Senegal show relatively strong and balanced governance profiles, particularly in Stability and Effectiveness. Conversely, Nigeria and Ghana exhibit weaker governance, especially in Rule of Law and Control of Corruption.

The bar chart (**Figure 13**) plots FDI inflows against the Control of Corruption Estimate, a World Bank governance metric that reflects perceptions of public sector corruption. While a positive trend might be expected—where countries with stronger control over corruption

High-tech Exports vs Trade (% of GDP) - 2023

The bubbles size denotes the percentage of the population with access to internet: Countries with higher tech exports appears to have more access to internet.

And also countries that are open for Trade (high GDP from Trade) are not necessarily Exporting High-tech.

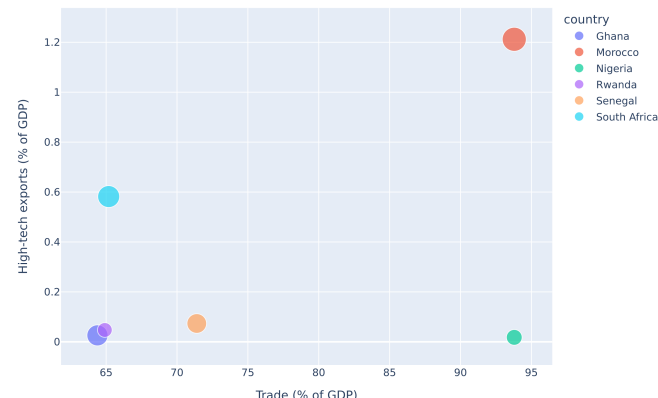


Figure 10. High-tech Exports vs Trade (% of GDP) - 2023.

Sectoral Composition Over Time - Nigeria

Nigeria is one of the countries with a very high percentage of the population (more than 60%) having no access to internet: The GDP generated from tech have been consistently negligible compared to other sectors of the economy.

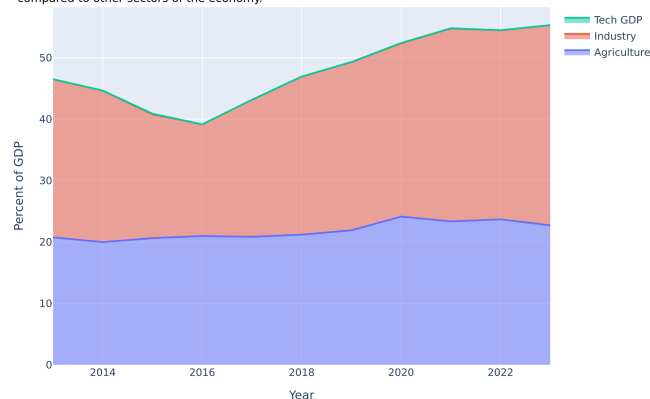


Figure 11. Sectoral Composition Over Time - Nigeria.

attract more investment—the data suggests a more complex reality. For instance, Senegal and Rwanda achieves relatively high FDI levels despite only moderate corruption control scores, indicating that other factors like regulatory efficiency or stability may also influence investor decisions.

Overall, these figures indicate that institutional strength matters—not just in terms of corruption control, but also through broader indicators like political stability, legal integrity, and government effectiveness. Countries looking to attract sustainable FDI should pursue comprehensive governance reforms, rather than focusing solely on anti-corruption. Strengthening public institutions, ensuring fair legal processes, and improving transparency can collectively create a more favorable investment climate.

7. Conclusion

This report has examined the development dynamics of six African countries through a multidimensional lens, revealing the complex and interconnected nature of progress across the continent. From economic resilience and human capital to governance and technology, the findings show that sustainable development is not driven by isolated gains, but by the alignment of policy, investment, and institutional strength.

Economic resilience post-COVID highlights how diversified economies and remittance flows can buffer shocks, though GDP growth alone does not guarantee job creation. In education,

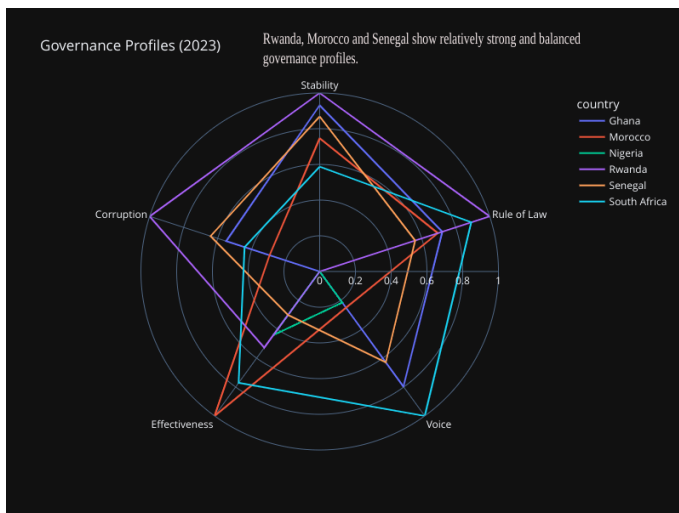


Figure 12. Governance Profiles (2023).

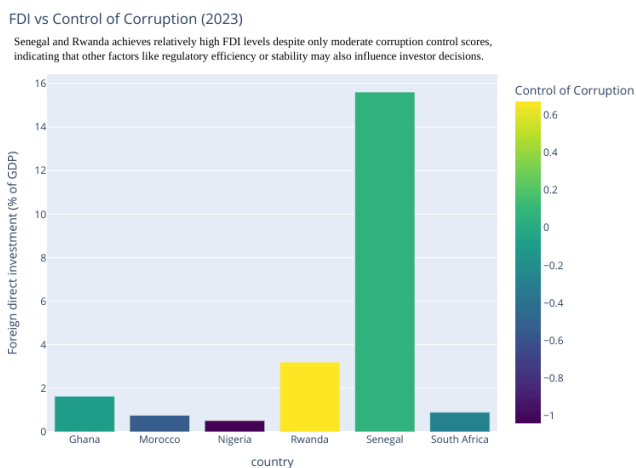


Figure 13. FDI vs Control of Corruption (2023).

we see that higher spending doesn't always equal better outcomes—emphasizing the need for holistic reforms. The health vs. wealth theme reinforces that investment in healthcare significantly improves outcomes like life expectancy and infant mortality, yet broader societal factors still matter. Technology and trade reveal a growing link between education and digital inclusion, but also expose the limited role of high-tech in many economies, notably Nigeria. Finally, governance quality emerges as a foundational pillar: countries with stronger institutions, even amid modest corruption controls, are better positioned to attract foreign investment and foster long-term stability.

Together, these insights suggest that Africa's development journey must be pursued through integrated, data-informed strategies—where economic, social, and institutional reforms move in tandem. This holistic approach is essential not just for resilience, but for inclusive and sustained transformation.

8. References

[1]"<https://databank.worldbank.org/reports.aspx?source=world-development-indicators>"